

# Eight pairs, three experiments

Each experiment trained eight pairs of agents. In a pair, both agents come from the same seed and the same code; one sees the road ahead and one does not. Each dot is one agent, each vertical line is one pair, and each experiment's verdict is quoted beside it from the file where it was fixed before the agents trained.

|                      |                                                                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| NOT SIGNIFICANT      | Phase D: eight pairs on a fixed climb<br>not significant · inconclusive (post-hoc) · the "blind" arm is not blind · C1 agents                          |
| INCONCLUSIVE         | Phase D2: eight pairs on a randomised climb<br>inconclusive · C1 agents, 50 000 steps                                                                  |
| SMALLER THAN THE MEI | C4: Phase D2's design, trained to 300 000 steps<br>smaller than the MEI (50) at 300 000 steps · one seed wide · the two tests disagree · not converged |

A **blue** dot is the sighted agent and an **orange** dot the blind one. When orange sits higher, the blind agent took more damage: preview helped in that pair. The row under each chart is blind – sighted. The team set 50 damage units in advance as the smallest effect worth caring about; the black bar beside each chart is that length.

The grey line is the current-grade rule, the best hand-written policy. Both agents beating it is a separate claim from preview (AUDIT.md C3): the blind agents beat it too.

Each experiment opens with every agent measured against the baseline ECU, as the result files print it: the share of the baseline's damage removed. That is learned protection. The pair chart under it is the preview test the team registered, in damage units.

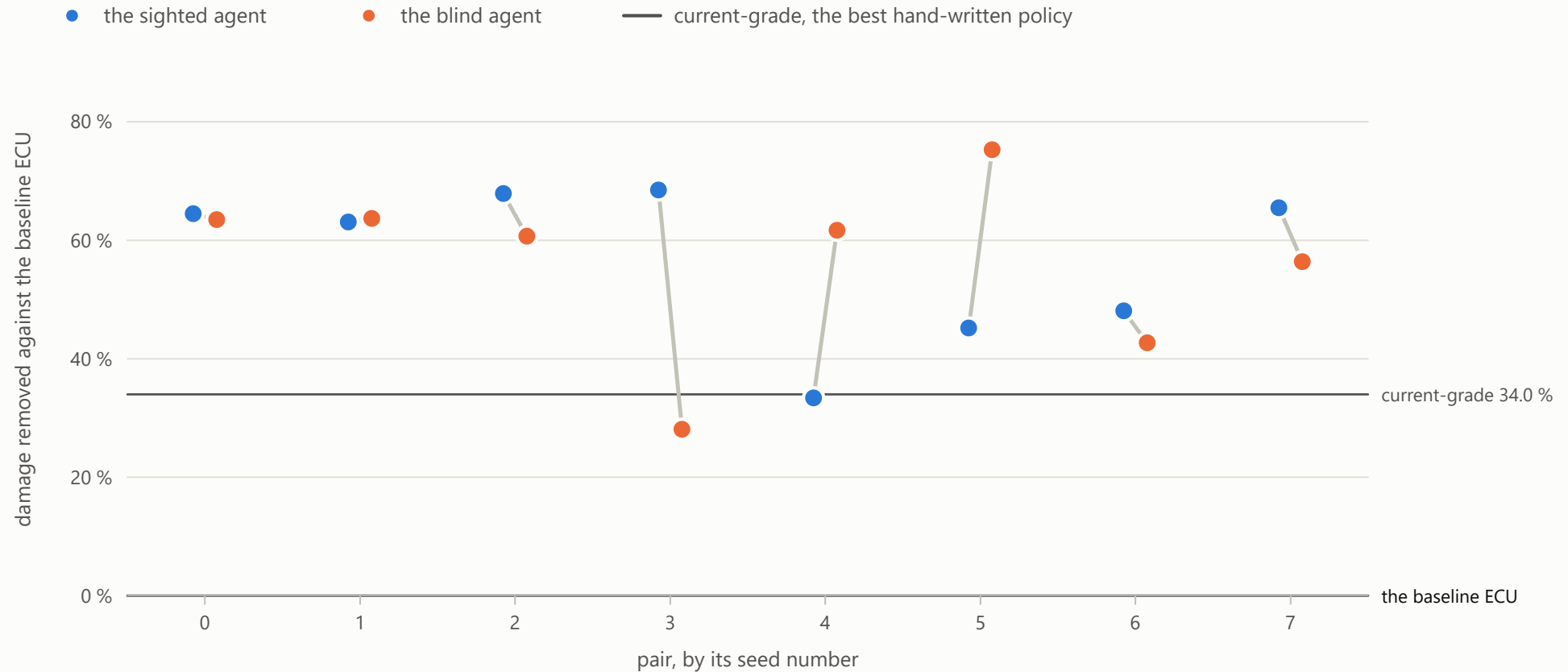
**What may be said, and what may not.** The sentences that may be said are the preregistered readings quoted under each chart. "Preview does not help" is not one of them: no experiment here supports it (CLAUDE.md).

## Phase D: every agent against the baseline ECU

share of the baseline ECU's median damage (959.8) that each agent removes · median of the twenty frozen episodes · higher is better

### SUPERVISION

learned protection, not preview: the blind agent is measured the same way



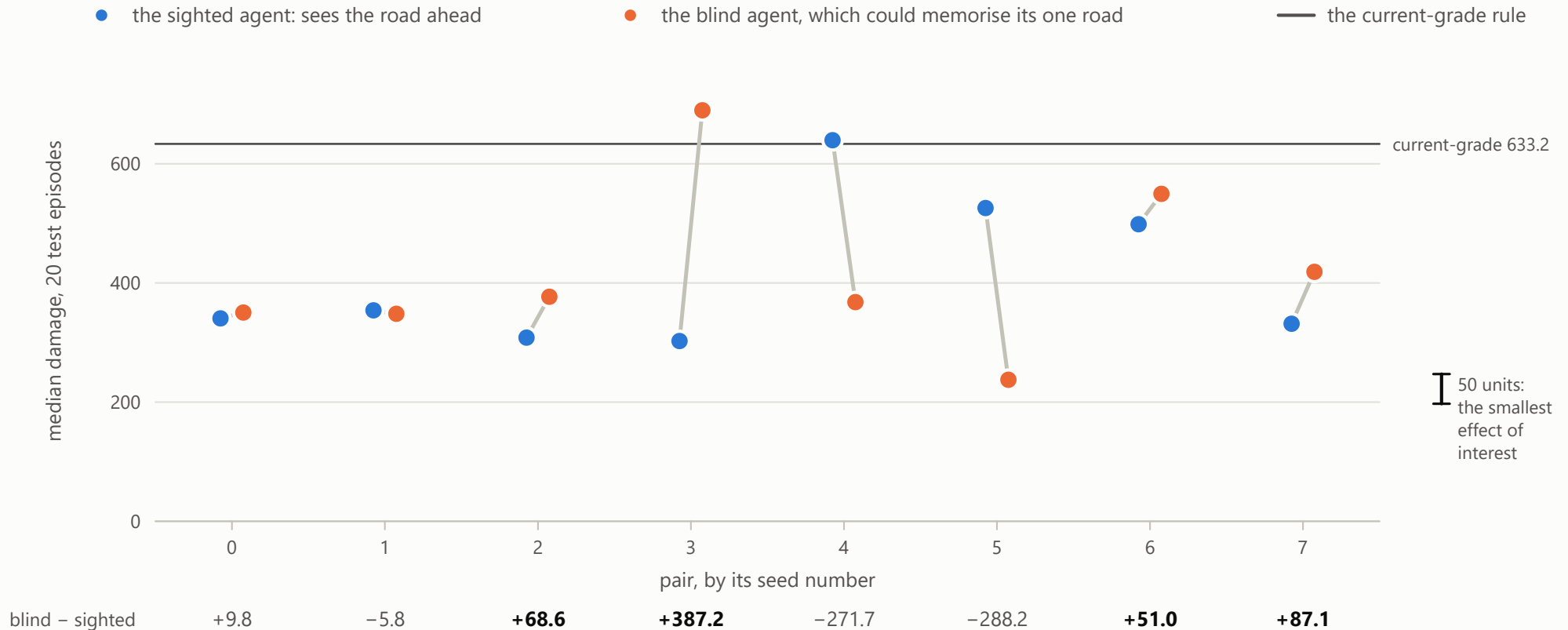
Read from the 'cuts median damage' lines evaluate.py printed in results/phase\_d\_seed\*.txt, each checked against the medians beside it. The grey line is current-grade, the best hand-written policy, and it is what an agent has to beat. The baseline ECU never enriches on this scenario and the agents can, so part of every cut measured against the baseline comes from a lever it does not pull here (CLAUDE.md, limitations). This chart measures learned protection; whether seeing the road ahead helps is the next chart's question, in the damage units its test was registered in.

## Phase D: eight pairs on a fixed climb

not significant · inconclusive (post-hoc) · the "blind" arm is not blind · C1 agents

**NOT SIGNIFICANT**

C1 budget, 50 000 training steps · 21-22 September 2026



A pair is two agents trained from the same seed, with the same code and budget; only the preview channel differs. Each dot is one agent's median damage over the same 20 frozen test episodes (evaluate.py); lower is less damage. blind – sighted above zero means the blind agent took more damage, so preview helped in that pair; bold marks a pair at or above the 50-unit minimum effect of interest. Phase D's blind arm is not blind: the climb comes at the same second on every episode, so the blind agent could learn when it comes (results/PREREGISTRATION.md limit 7). Its minimum effect of interest was set after this result, so reading it against 50 units is post-hoc. The unprotected baseline ECU scores 959.8 here, off the top of the chart. Data: results/phase\_d\_seed\*.txt.

**NOT SIGNIFICANT**

Not statistically significant, with agents at the C1 budget

## INCONCLUSIVE (POST-HOC)

Inconclusive, and a post-hoc reading: the MEI (50 units) was set after Phase D's result was known, so this classification was not preregistered. The experiment cannot tell "no effect" from "an effect the team cares about"

RESULT: NOT SIGNIFICANT at alpha 0.05.

These are C1 agents: 50 000 steps, 11 training episodes each, which train.py itself calls 'the first bad run'. So the claim is 'with agents trained to the C1 budget, preview does not separate from seed noise' -- NOT 'preview does not help'. See results/PREREGISTRATION.md limit 6.

results/PHASE\_D\_RESULT.txt:26

AND THE BLINDED ARM IS NOT BLIND. The road is the same hill at the same second in every episode, and the blind agent's thermal state takes a distinct value at every step -- a clock on a road it can memorise. So this compares an explicit preview channel with an implicit one, not foresight with none. Four explanations for the null are live and this experiment separates none of them. Randomise the climb per episode before reading anything into it. See results/PREREGISTRATION.md limits 7 and 8.

results/PHASE\_D\_RESULT.txt:33

RESULT: INCONCLUSIVE [MEI set AFTER this result -- see below]  
This experiment cannot tell 'no effect' from 'an effect the team would care about'. It is a statement about the experiment's size, not a statement that preview does nothing.  
power\_analysis.py explains it -- but was written after this result, so it is an explanation, not a prediction.

Phase D's primary test is unaffected: its verdict was 'not significant' and still is. The MEI was set on 22 September, AFTER Phase D's result was known, so Phase D's classification under the MEI rule is a POST-HOC reading and is labelled so.  
results/PREREGISTRATION.md section 5 records the ordering.

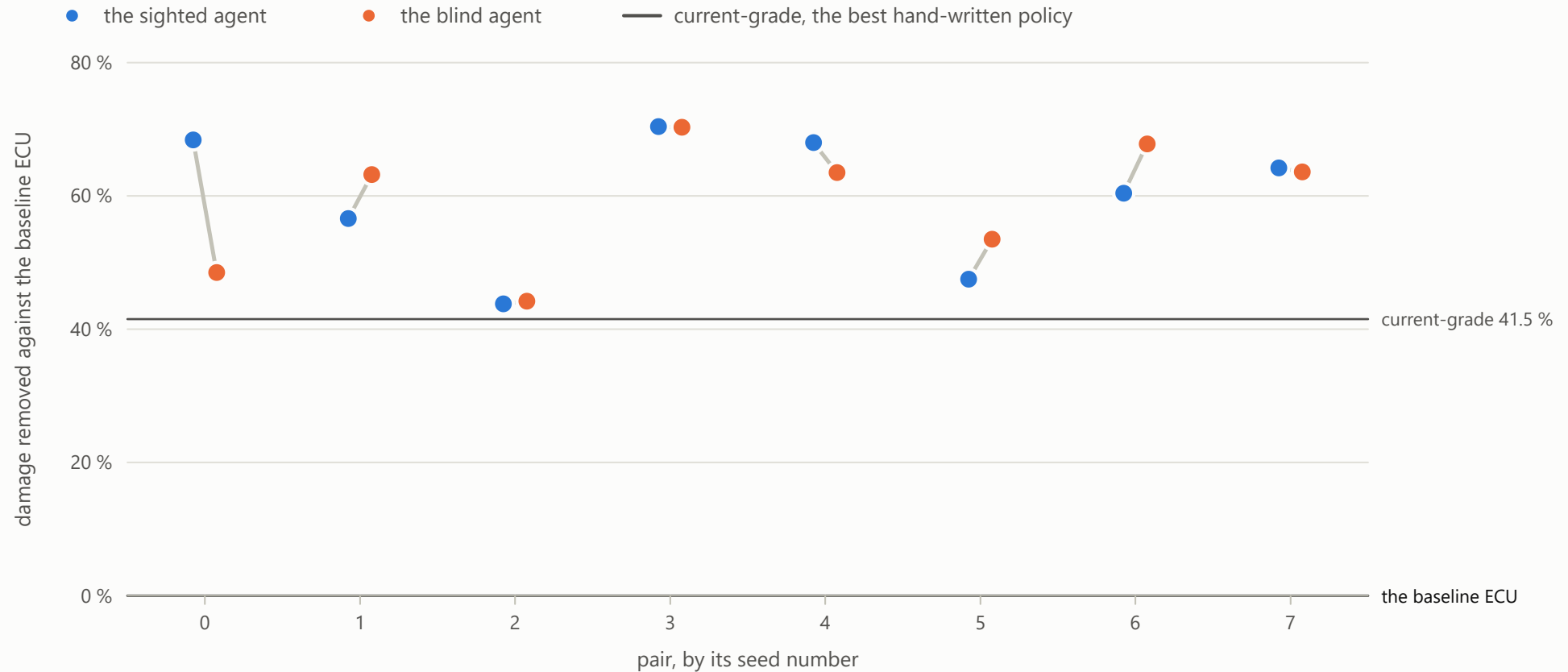
results/PHASE\_D2\_RESULT.txt:65

## Phase D2: every agent against the baseline ECU

share of the baseline ECU's median damage (1118.0) that each agent removes · median of the twenty frozen episodes · higher is better

### SUPERVISION

learned protection, not preview: the blind agent is measured the same way



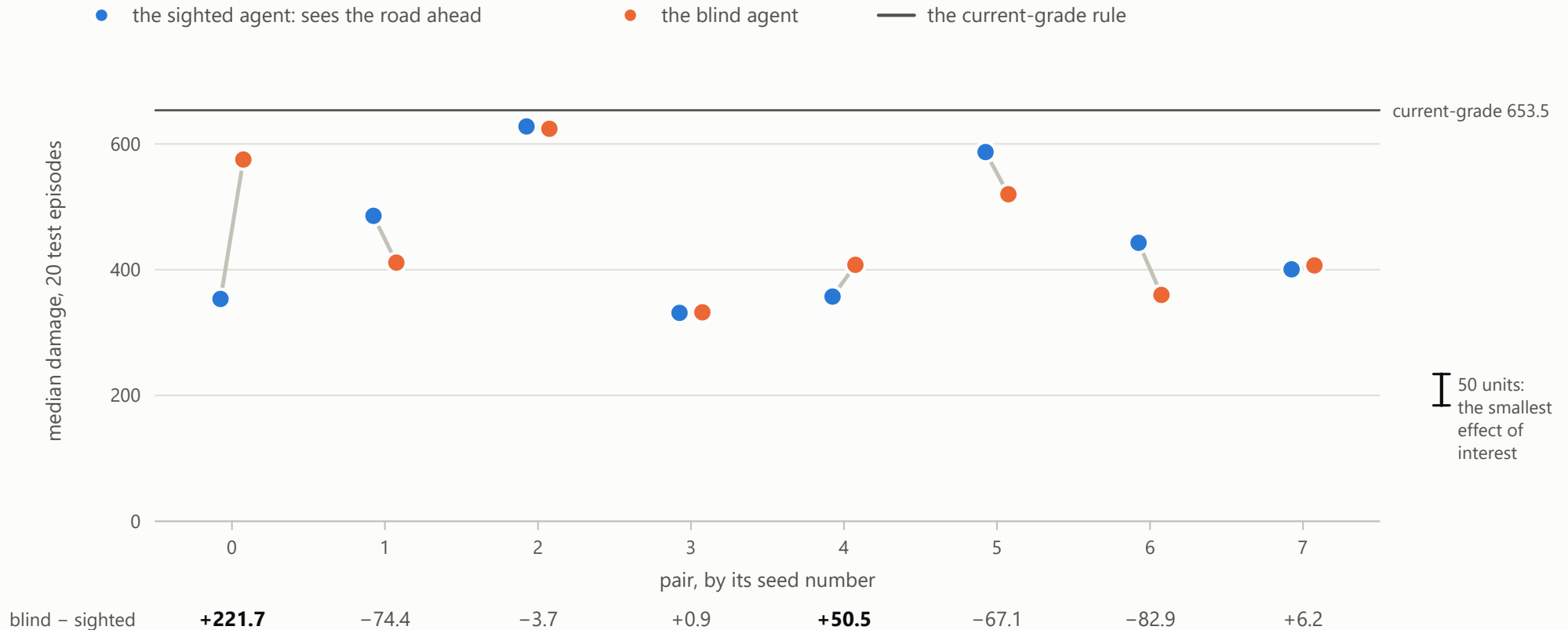
Read from the 'cuts median damage' lines evaluate.py printed in results/d2\_seed\*.txt, each checked against the medians beside it. The grey line is current-grade, the best hand-written policy, and it is what an agent has to beat. The baseline ECU never enriches on this scenario and the agents can, so part of every cut measured against the baseline comes from a lever it does not pull here (CLAUDE.md, limitations). This chart measures learned protection; whether seeing the road ahead helps is the next chart's question, in the damage units its test was registered in.

## Phase D2: eight pairs on a randomised climb

inconclusive · C1 agents, 50 000 steps

**INCONCLUSIVE**

C1 budget, 50 000 training steps · 22-23 September 2026



A pair is two agents trained from the same seed, with the same code and budget; only the preview channel differs. Each dot is one agent's median damage over the same 20 frozen test episodes (evaluate.py); lower is less damage. blind – sighted above zero means the blind agent took more damage, so preview helped in that pair; bold marks a pair at or above the 50-unit minimum effect of interest. The climb's start and grade were drawn per episode and the blind arm was checked blind before training (check\_random\_road.py). The minimum effect of interest, 50 units, was set before this experiment. The unprotected baseline ECU scores 1118.0 here, off the top of the chart. Data: results/d2\_seed\*.txt.

**INCONCLUSIVE**

Inconclusive: the experiment cannot tell "no effect" from "an effect the team cares about"

RESULT: INCONCLUSIVE

This experiment cannot tell 'no effect' from 'an effect the team would care about'. It is a statement about the experiment's size, not a statement that preview does nothing.

power\_analysis.py predicted this cell BEFORE this experiment ran, from Phase D's measured spread.

results/PHASE\_D2\_RESULT.txt:30

Both are C1 agents -- 50 000 steps, 11 training episodes each, which train.py calls 'the first bad run'. A null from either says 'with agents trained to the C1 budget', never 'preview does not help'. The next suspect after the design flaw is the budget (C4).

results/PHASE\_D2\_RESULT.txt:86

## C4: every agent against the baseline ECU

share of the baseline ECU's median damage (1118.0) that each agent removes · median of the twenty frozen episodes · higher is better

### SUPERVISION

learned protection, not preview: the blind agent is measured the same way



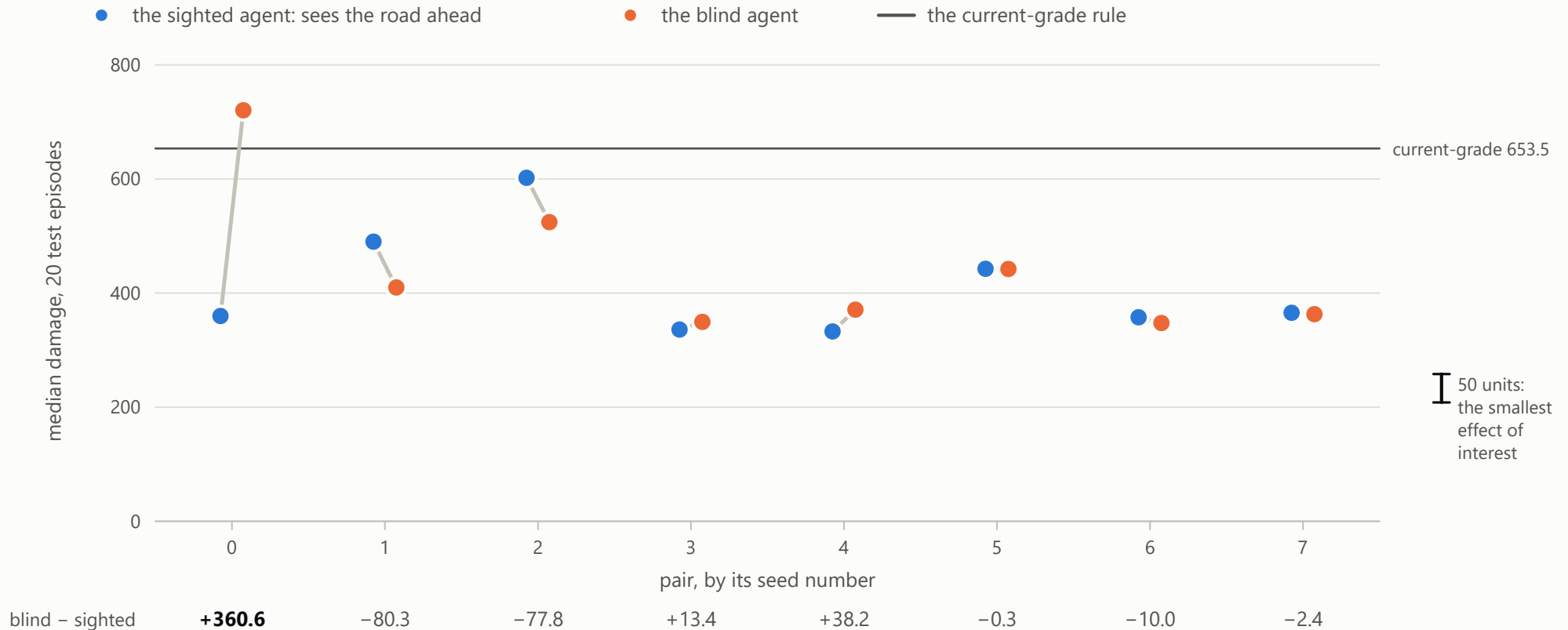
Read from the 'cuts median damage' lines evaluate.py printed in results/c4\_seed\*.txt, each checked against the medians beside it. The grey line is current-grade, the best hand-written policy, and it is what an agent has to beat. The baseline ECU never enriches on this scenario and the agents can, so part of every cut measured against the baseline comes from a lever it does not pull here (CLAUDE.md, limitations). This chart measures learned protection; whether seeing the road ahead helps is the next chart's question, in the damage units its test was registered in.

## C4: Phase D2's design, trained to 300 000 steps

smaller than the MEI (50) at 300 000 steps · one seed wide · the two tests disagree · not converged

**SMALLER THAN THE MEI**

C4 budget, 300 000 training steps, not converged · 23-24 September 2026



A pair is two agents trained from the same seed, with the same code and budget; only the preview channel differs. Each dot is one agent's median damage over the same 20 frozen test episodes (evaluate.py); lower is less damage. blind – sighted above zero means the blind agent took more damage, so preview helped in that pair; bold marks a pair at or above the 50-unit minimum effect of interest. C4's agents are Phase D2's agents trained longer, not a replication: every one retraced its D2 twin bit for bit through 50 000 steps. By the preregistered rule they had not converged (results/C4\_RESULT.txt). The unprotected baseline ECU scores 1118.0 here, off the top of the chart. Data: results/c4\_seed\*.txt.

**SMALLER THAN THE MEI**

Preview's effect is below 50 damage units, a threshold the team set in advance, at 300 000 training steps, on a road with one grade change per episode. Training had not settled, and the result hangs on one seed: if one seed flipped, it would be inconclusive.

NOT-CONVERGED

Training had not settled

RESULT: SMALLER THAN THE MEI

results/C4\_RESULT.txt:42

H1: the effect is smaller than the MEI (50 units)

exact one-sided sign test      p = 0.0352      (7 of 8 seeds below 50)

exact paired permutation test      p = 0.3867      (sensitivity)

results/C4\_RESULT.txt:28

THE TWO TESTS DISAGREE on 'smaller than the MEI'. Both reported;  
the sign test is primary.

results/C4\_RESULT.txt:33

NOT-CONVERGED sighted 2/8 blind 1/8 pairs 1/8 band 25 need 6 and 7 runs runs\_c4

results/C4\_RESULT.txt:48

THE READING, as declared before any C4 agent trained (section 2a):

Preview's effect is below the MEI at 300 000 steps: (i) is supported  
AT THIS BUDGET. The agents were still changing, so (ii) is not ruled  
out.

results/C4\_RESULT.txt:50

| (i) | preview genuinely buys little at this configuration | tested jointly with (ii) – see 2a |  
| (ii) | **the agents are undertrained** – C1, 50 000 steps, 11 training episodes | **yes – this is the one thing C4 changes** |  
| (iii) | the preview carries little: one grade step per episode | a special case of (i) at this configuration. C4 inherits D2's road  
unchanged, so any (i) finding reads "with one grade step per episode" |

results/PREREGISTRATION\_C4.md:41

2. **It rests on the sign test's threshold, and the sensitivity test does not agree.** 7 of 8 is exactly the count the sign test needs at eight seeds; one seed the other way and the cell would be INCONCLUSIVE. The permutation test, which weighs magnitudes, does not reject (p = 0.3867), because one seed carries a large effect the other way: in seed 0 the blinded agent took

720.5 against the sighted agent's 359.9, a difference of +360.6, and that one value lifts the mean to +30.2. Section 5 fixed the rule for this before the run: both are reported, the sign test is primary, and neither is chosen after the fact. \*\*So the result is: most seeds show no benefit from preview, one shows a large one, and whether the average effect is below 50 is not settled.\*\*

results/PREREGISTRATION\_C4.md:633

# The numbers behind each chart

## Phase D: every agent against the baseline ECU

| seed | sighted removes, % | blind removes, % | current-grade removes, % |
|------|--------------------|------------------|--------------------------|
| 0    | 64.5               | 63.5             | 34.0                     |
| 1    | 63.1               | 63.7             | 34.0                     |
| 2    | 67.9               | 60.7             | 34.0                     |
| 3    | 68.5               | 28.1             | 34.0                     |
| 4    | 33.4               | 61.7             | 34.0                     |
| 5    | 45.2               | 75.3             | 34.0                     |
| 6    | 48.1               | 42.7             | 34.0                     |
| 7    | 65.5               | 56.4             | 34.0                     |

## Phase D: eight pairs on a fixed climb

| seed | baseline ECU | current-grade | sighted | blind | blind – sighted |
|------|--------------|---------------|---------|-------|-----------------|
| 0    | 959.8        | 633.2         | 340.5   | 350.3 | +9.8            |
| 1    | 959.8        | 633.2         | 354.0   | 348.2 | −5.8            |
| 2    | 959.8        | 633.2         | 308.2   | 376.8 | +68.6           |
| 3    | 959.8        | 633.2         | 302.5   | 689.7 | +387.2          |
| 4    | 959.8        | 633.2         | 639.5   | 367.8 | −271.7          |
| 5    | 959.8        | 633.2         | 525.7   | 237.5 | −288.2          |
| 6    | 959.8        | 633.2         | 498.5   | 549.5 | +51.0           |
| 7    | 959.8        | 633.2         | 331.5   | 418.6 | +87.1           |

## Phase D2: every agent against the baseline ECU

| seed | sighted removes, % | blind removes, % | current-grade removes, % |
|------|--------------------|------------------|--------------------------|
| 0    | 68.4               | 48.5             | 41.5                     |

|   |      |      |      |
|---|------|------|------|
| 1 | 56.6 | 63.2 | 41.5 |
| 2 | 43.8 | 44.2 | 41.5 |
| 3 | 70.4 | 70.3 | 41.5 |
| 4 | 68.0 | 63.5 | 41.5 |
| 5 | 47.5 | 53.5 | 41.5 |
| 6 | 60.4 | 67.8 | 41.5 |
| 7 | 64.2 | 63.6 | 41.5 |

#### Phase D2: eight pairs on a randomised climb

| seed | baseline ECU | current-grade | sighted | blind | blind – sighted |
|------|--------------|---------------|---------|-------|-----------------|
| 0    | 1118.0       | 653.5         | 353.5   | 575.2 | +221.7          |
| 1    | 1118.0       | 653.5         | 485.7   | 411.3 | −74.4           |
| 2    | 1118.0       | 653.5         | 627.8   | 624.1 | −3.7            |
| 3    | 1118.0       | 653.5         | 331.4   | 332.3 | +0.9            |
| 4    | 1118.0       | 653.5         | 357.3   | 407.8 | +50.5           |
| 5    | 1118.0       | 653.5         | 587.1   | 520.0 | −67.1           |
| 6    | 1118.0       | 653.5         | 442.8   | 359.9 | −82.9           |
| 7    | 1118.0       | 653.5         | 400.6   | 406.8 | +6.2            |

#### C4: every agent against the baseline ECU

| seed | sighted removes, % | blind removes, % | current-grade removes, % |
|------|--------------------|------------------|--------------------------|
| 0    | 67.8               | 35.6             | 41.5                     |
| 1    | 56.2               | 63.3             | 41.5                     |
| 2    | 46.1               | 53.1             | 41.5                     |
| 3    | 69.9               | 68.7             | 41.5                     |
| 4    | 70.2               | 66.8             | 41.5                     |

|   |      |      |      |
|---|------|------|------|
| 5 | 60.4 | 60.4 | 41.5 |
| 6 | 68.0 | 68.9 | 41.5 |
| 7 | 67.3 | 67.5 | 41.5 |

#### C4: Phase D2's design, trained to 300 000 steps

| seed | baseline ECU | current-grade | sighted | blind | blind – sighted |
|------|--------------|---------------|---------|-------|-----------------|
| 0    | 1118.0       | 653.5         | 359.9   | 720.5 | +360.6          |
| 1    | 1118.0       | 653.5         | 490.2   | 409.9 | −80.3           |
| 2    | 1118.0       | 653.5         | 602.2   | 524.4 | −77.8           |
| 3    | 1118.0       | 653.5         | 336.1   | 349.5 | +13.4           |
| 4    | 1118.0       | 653.5         | 332.8   | 371.0 | +38.2           |
| 5    | 1118.0       | 653.5         | 442.6   | 442.3 | −0.3            |
| 6    | 1118.0       | 653.5         | 357.6   | 347.6 | −10.0           |
| 7    | 1118.0       | 653.5         | 365.5   | 363.1 | −2.4            |

Built by `plot_agent_pairs.py`. Damage figures are read from `results/<experiment>_seed<k>.txt` by the analysis scripts' own reader, and every verdict line is quoted by `app/agent_catalog.py`; nothing on this page is computed. Every value is printed in `figures/agent_pairs/numbers.txt`.